What happens when coupling switches on?

exp085 · 17 August 2026 · Draft

Abstract

Two slightly detuned PING circuits begin from the same mature uncoupled trajectory before reciprocal excitation switches on. Across 11 coupling strengths, the unwrapped relative phase changes from a persistent drift of 30.1 rad/s to −0.01 rad/s. The experiment uses runtime-state branching to isolate coupling onset from network activation and shows the transition directly without imposing a binary synchronization threshold.

Methods

1. **Equilibrate two detuned circuits.** Independent 100 Hz/channel Poisson streams drive two 80 E / 20 I PING components. Circuit A uses 4 ms inhibitory decay and circuit B uses 5 ms. Both run uncoupled for 1000 ms.
2. **Branch one mature state.** The executor captures voltages, refractory counters, conductances, population histories, and input-delay histories at coupling onset. That identical state branches across 11 reciprocal coupling strengths. Private future inputs remain paired across K and independent between circuits.
3. **Measure relative phase.** E-population rates are smoothed with a 5 ms Gaussian kernel and filtered over 20–60 Hz. Hilbert phases define the unwrapped A-minus-B relative phase, expressed as its change from coupling onset. Figures divide this phase change by 2π to report relative cycles gained. The final 100 ms are excluded to prevent the finite-record filter boundary from appearing as a phase jump. This continuous trajectory is the primary result; no locking threshold or latency is inferred.
4. **Measure supporting responses.** The registered `exp085-dominant-rhythm-v1` policy reports each circuit's dominant rhythm from the final 1.5 s. Frequency difference, descriptive phase-locking value, and population firing rates support the phase trajectories.

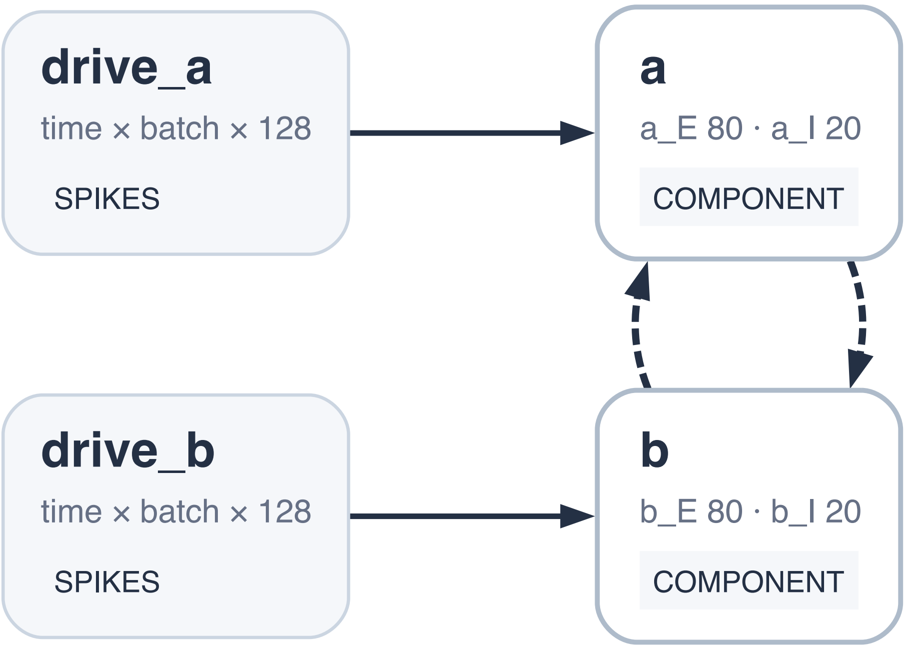

Figure 1: Two-circuit coupling-onset graph. Independent spike inputs drive circuits A and B. Four reciprocal AMPA projections share coupling strength K; their weights are zero during equilibration and vary only after the common runtime-state branch. Local PING topology and every non-coupling parameter remain fixed.

Results

The relative-phase trajectories expose the response to coupling without reducing it to a label. At K = 0 the circuits repeatedly lap one another. Increasing K changes the slope and structure of that drift; the strongest condition ends with a median phase-locking value of 0.99.

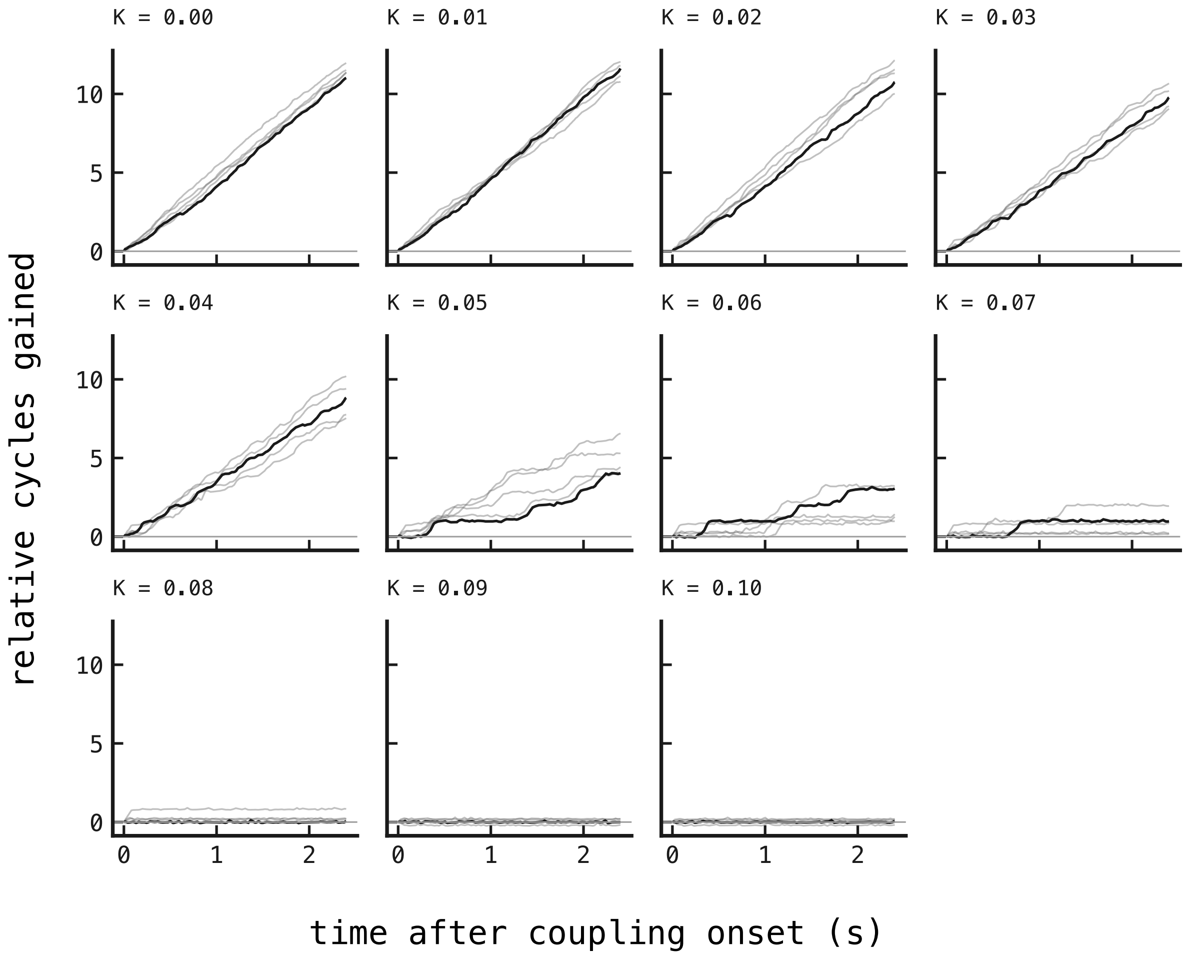

Figure 2: Cumulative A-minus-B relative cycles gained after coupling onset. One vertical unit means that A has completed one additional cycle relative to B. Panels increase in reciprocal coupling K from left to right and top to bottom. Every trace is zeroed at onset; the final 100 ms are excluded as a registered filter-edge guard. The black trace is one predeclared paired trial; grey traces are the other 4 trials from the same mature-state and input protocol. Persistent slope indicates repeated lapping, plateaus indicate transient capture, and a bounded trace indicates a stable phase relationship.

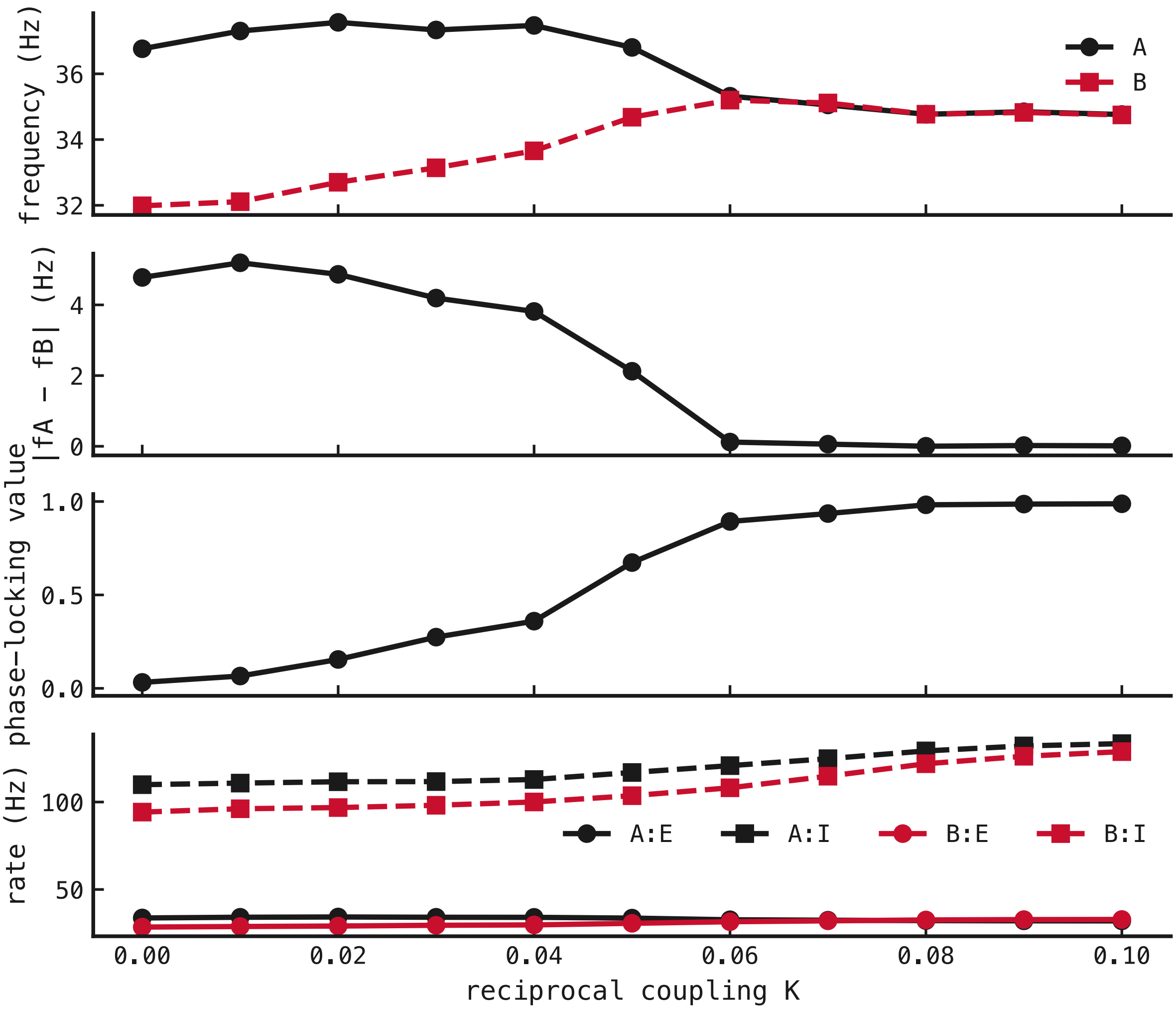

Figure 3: Supporting coupling response. First: median dominant E-population frequencies of circuits A and B in hertz. Second: their absolute difference in hertz. Third: median descriptive phase-locking value across 5 paired trials. Fourth: mean E and I firing rates in hertz for both circuits. K is the shared weight of the four reciprocal AMPA projections.

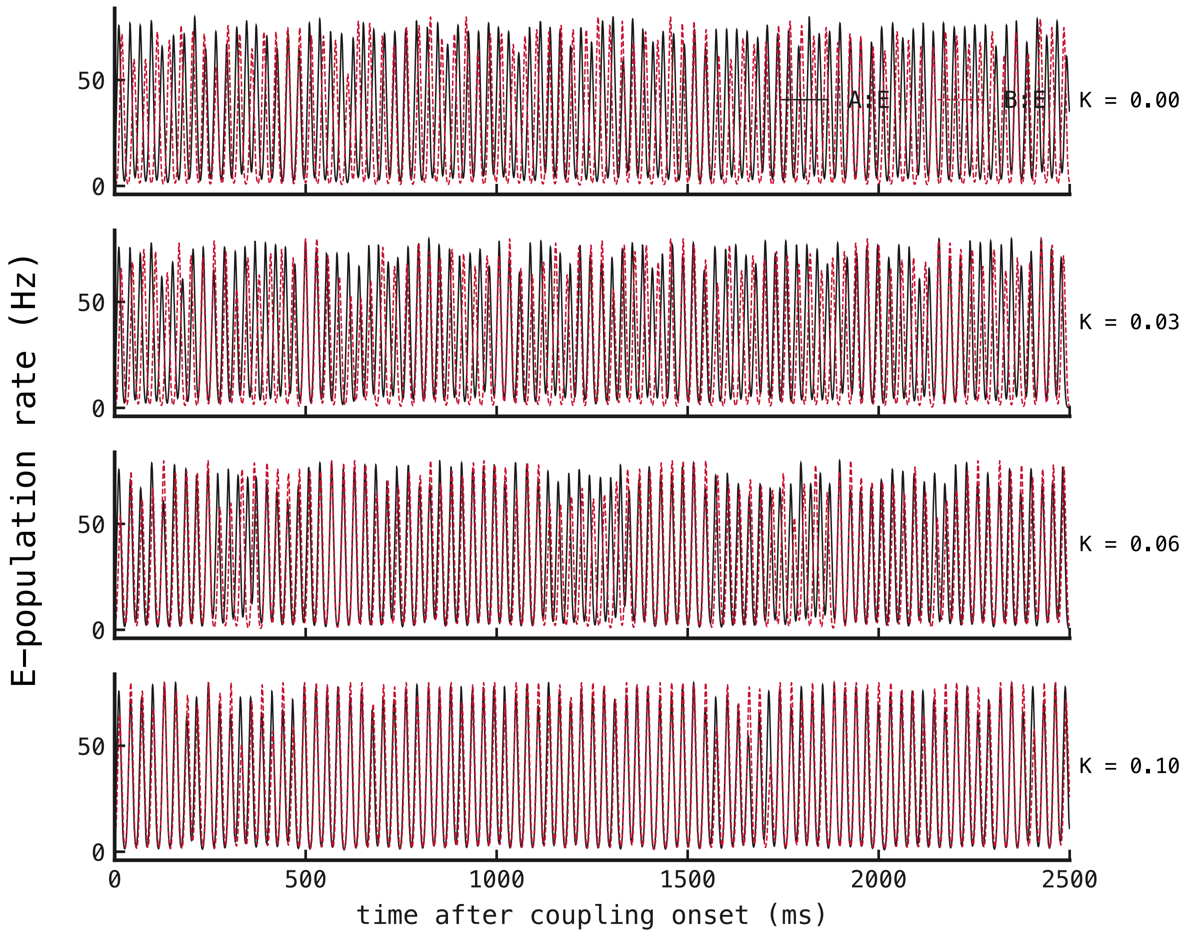

Figure 4: Excitatory population activity after coupling onset at the four predeclared K values. Time is in milliseconds and rate is in hertz after the registered 5 ms smoothing. Circuit A is black and circuit B is dashed red; all panels use the same paired trial and axes. The sequence makes repeated lapping, transient alignment, and shared volley timing directly comparable.